THE ENGINE SERIES · No.01

BUILD A BIGGER ENGINE

The science-backed guide to raising your **VO_2max** for running and hybrid performance. What the latest research actually says, written so you can use it on Monday morning.

Built for runners, Hyrox athletes and anyone chasing a stronger, healthier engine.

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START HERE

YOUR ENGINE IS TRAINABLE. LET'S BUILD IT.

Your **VO_2max** is the size of your aerobic engine: the most oxygen your body can take in, deliver and burn when you are working flat out. It sets the ceiling on your endurance, and it is one of the strongest predictors of long term health we have ever measured. The good news is simple. It responds to training, often dramatically, at almost any age.

This guide pulls together the most current and reliable sports science on raising VO_2max, then translates it into plain language and clear actions. No jargon for its own sake. No magic shortcuts. Just the levers that genuinely move the needle, ranked by how much they matter, with honest notes on what is overhyped.

Whether you are training for a first 10k, a marathon, or your next Hyrox, the engine you build is the same. Here is how to make it bigger.

> **HOW TO READ THIS GUIDE**
>
> Each section ends with a **Takeaway** (the one thing to remember) and many include a **Try This** box you can act on today. Skim the boxes if you are short on time, then come back for the science.

What's inside

01 THE THREE PILLARS

It is tempting to treat VO_2max as the only number that matters. It is not. Endurance performance rests on **three pillars**, and the fastest athlete is rarely the one with the single highest score in any one of them. They are the one with the best combination.

Pillar 1: VO_2max, your ceiling

This is the maximum oxygen your body can use per minute, usually given in millilitres per kilogram of body weight per minute. Think of it as the size of your engine. A bigger engine gives you more room at the top. The American Heart Association now treats VO_2max as a clinical vital sign, and a low score predicts mortality more strongly than smoking, high blood pressure or type 2 diabetes. This is as much a health document as a performance one.

Pillar 2: Lactate threshold, your usable share

Your threshold decides how much of that ceiling you can actually hold for a long time. Two runners can share the same VO_2max, but the one who can cruise at 88 percent of it will crush the one stuck at 75 percent. Threshold is highly trainable, and as you will see, it is the centrepiece of the most talked about training method in distance running right now.

Pillar 3: Running economy, your efficiency

Economy is how much oxygen you burn to hold a given pace. A more economical runner uses less fuel at the same speed, which is exactly why an athlete with a lower VO_2max can still beat one with a higher ceiling. Economy is the quiet pillar, and strength training is one of the best ways to improve it (more on that in Section 6).

> **A bigger engine raises your ceiling. Threshold and economy decide how much of it you get to use.**

THE TAKEAWAY

Chase all three pillars, not just VO_2max. The training in this guide is built to lift your ceiling **and** raise the share of it you can sustain.

02 CAN ANYONE IMPROVE?

This is the question that quietly worries a lot of people: **what if I am just not built for it?** Here is the honest, encouraging answer from the research.

Genetics do play a real role. Studies suggest roughly half of your untrained, baseline VO_2max is inherited, along with about a quarter of how much you can improve.[1] The famous HERITAGE Family Study found that after a standard 20 week programme, gains varied enormously from person to person, and somewhere around 10 to 20 percent of people barely responded at all to that particular cookie cutter plan.[2]

So far that might sound discouraging. It is not, and here is why.

"Non-response" is usually a dose problem, not a you problem

The crucial detail is that those low responders were following one fixed, moderate programme. When researchers **increase the dose**, by adding intensity or volume, much of that non-response disappears.[3] In other words, a stubborn engine often just needs a stronger or different signal. The studies that used proper high intensity intervals saw meaningful gains across the board, with average improvements noticeably larger than gentle steady state plans produced.[4]

A second lever is variety. If one type of training is not working for you, switching the type of stimulus, for example moving from easy continuous runs to intervals, or adding strength work, can unlock progress that the original plan never would.[3]

~50%
of baseline VO_2max is influenced by genetics

~25%
of your trainability is inherited, the rest is up to you

0.5L
typical per minute VO_2max gain from interval based plans

THE TAKEAWAY

Almost everyone can raise their VO_2max meaningfully. If progress stalls, the answer is rarely "bad genetics". It is usually **more intensity, more volume, or a different stimulus**. Your job is to find the dose your engine responds to.

03 THE ENGINE ROOM

To train VO_2max well, it helps to know what you are actually changing. Improvements come from two places, and they run on very different clocks.

Central adaptations: a stronger pump

These happen in your heart and circulation. The headline change is a bigger **stroke volume**, meaning your heart pushes more blood with every beat. Your blood and plasma volume expand too, so there is simply more oxygen carrying fluid in the system. More blood per beat plus more beats worth of capacity equals more oxygen delivered to working muscle. This is the largest single driver of VO_2max, and the encouraging part is that it moves relatively quickly.

Peripheral adaptations: hungrier muscles

These happen inside the muscle itself: more **capillaries** threading oxygen to the fibres, more **mitochondria** (the cellular engines that actually use the oxygen), and more of the enzymes that power aerobic metabolism. These changes improve how much oxygen you extract and how efficiently you use it. They build more slowly, but they keep building for years.

The timeline that should keep you patient and consistent

- **Weeks 6 to 12:** Most of the central, heart driven gains appear. This is why a focused block can lift your VO_2max noticeably in a couple of months.
- **Months into years:** The capillary and mitochondrial changes keep accumulating. This is the deep, durable fitness that separates a third year runner from a first year one.

> **Your heart adapts in weeks. Your muscles keep adapting for years. Both reward showing up.**

THE TAKEAWAY

Expect early wins in the first two to three months, then trust the slow build. The athletes who win the long game are the ones who stay consistent long enough for the **peripheral adaptations** to compound.

04 INTERVALS DONE RIGHT

If you only change one thing, change this. **High intensity intervals are the most potent single tool for raising VO_2max**, and the research is now remarkably specific about how to structure them.

The magic happens above 90 percent

The goal of a VO_2max session is to spend as much total time as possible working at or above roughly 90 percent of your maximum. That is the zone that signals your body to build a bigger engine. Everything about good interval design comes back to maximising time in that zone without falling apart.

Longer reps beat short, sharp bursts

This surprises people. A 2024 study compared classic long intervals (four reps of three minutes) against a flurry of short ones (twenty one reps of thirty seconds) at the same intensity and total work. The long intervals produced clearly more time above 90 percent of VO_2max.[5] Short bursts feel hard, but they let your oxygen uptake sag during each brief effort. Longer reps hold you in the productive zone.

The numbers the latest meta-analyses keep landing on

A large 2025 network meta-analysis of 51 studies and over 1,200 athletes mapped out an inverted U shaped sweet spot. Benefits peaked at around **140 seconds of work** paired with a **work to recovery ratio near 0.85**, which in practice is roughly equal work and rest.[6] Separately, the body of evidence points to **three to five minute reps** as the range that delivers the biggest VO_2max improvements.[7]

LEVER	WHAT THE RESEARCH SUGGESTS
Rep length	3 to 5 minutes for the biggest VO_2max gains. A sweet spot around 140 seconds in pooled data.
Work to rest	Roughly 1:1 (a ratio near 0.85). A 4 minute rep pairs with about 4 to 5 minutes recovery.
Intensity	Hard enough to reach and hold above 90 percent of VO_2max. Think 3k to 5k race effort.
Recovery type	Keep moving. Active recovery keeps the system primed so you hit the zone faster next rep.
Total quality time	Aim to accumulate meaningful minutes above 90 percent, often 12 to 20 minutes across the session.

TRY THIS · A CLASSIC VO_2MAX SESSION

After a thorough 15 minute warm up:

- **5 × 3 minutes** at hard 3k to 5k effort
- **3 minutes** easy jogging between each (keep moving)
- Cool down 10 minutes easy

That is 15 minutes of high quality work in the engine building zone. Once or twice a week is plenty.

THE TAKEAWAY

Favour **3 to 5 minute reps** with roughly **equal recovery**, run hard enough to live above 90 percent, and **keep jogging** between reps. This single habit is your biggest lever.

05 TRAIN SMART

Intervals are powerful, but you cannot do them every day. How you arrange **all** of your training across the week, your intensity distribution, is what lets you absorb the hard work and keep improving instead of breaking down.

The three zones, simply

- **Zone 1, easy:** Below your first threshold. Conversational. You can do a lot of this with little fatigue.
- **Zone 2, the grey zone:** Moderately hard, between your two thresholds. Comfortable enough to overdo, hard enough to tire you.
- **Zone 3, hard:** Above your second threshold. This is where VO_2max work lives.

Polarized: the 80/20 rule

The most consistent finding in endurance research is that successful athletes spend most of their time easy and a smaller slice genuinely hard, while largely avoiding the grey zone. A **polarized** approach is often summarised as roughly 80 percent easy, 20 percent hard.[8] Keeping the easy days truly easy is what lets the hard days be hard enough to matter. Most amateurs make their easy runs too fast and their hard runs too soft, and get stuck in the middle.

Pyramidal: a close cousin

A **pyramidal** distribution keeps the big easy base but allows a meaningful amount of moderate threshold work, with a smaller cap of very hard work on top. It is often favoured by time crunched athletes and during race specific phases for the half marathon and marathon.

The Norwegian double threshold method

This is the most discussed idea in distance running today, popularised by the Ingebrigtsen brothers and triathlon champion Kristian Blummenfelt. The core idea: do a high **volume** of controlled threshold work, sometimes **two threshold sessions in a single day**, kept deliberately at a moderate lactate level of roughly 2.5 to 4 millimoles.[9] Because each session sits in a sustainable window rather than an all out grind, athletes pack in far more quality work near threshold without the deep fatigue and injury risk of constant maximal efforts. Over time that builds an enormous aerobic base, sharpened by one weekly dose of faster, race specific work.[9]

REALITY CHECK FOR THE REST OF US

The full method relies on lactate testing and twice daily training few amateurs can manage. The usable lesson is the principle, not the logistics: controlled, repeatable threshold work, which most people can approximate with an effort you could sustain for about an hour, roughly 85 percent of max heart rate or an RPE of 6 to 7 out of 10.[9]

TRY THIS · A CONTROLLED THRESHOLD SESSION

- **5 × 6 minutes** at "comfortably hard", an effort you could hold for about an hour
- **1 minute** easy jog between reps
- You should finish feeling you had a little more in the tank, not wrecked

THE TAKEAWAY

Make your easy days **genuinely easy** and protect a small dose of real intensity. Whether you lean polarized or pyramidal, controlled threshold volume plus a weekly VO_2max session is a proven recipe.

06 THE HYBRID EDGE

If you are a Hyrox or hybrid athlete, you face a question pure runners never have to: **how do I get a bigger engine and stay strong at the same time?** For years the fear was the "interference effect", the idea that cardio kills your gains. The modern evidence is far more reassuring.

Good news: strength and endurance mostly play nicely

The largest review on the topic, covering 43 studies, found that training strength and endurance together did **not** meaningfully reduce gains in maximal strength or muscle size across a wide range of people and methods.[10] The main casualty is **explosive** power, things like jump height and sprint speed, which can drop by around a quarter, especially when both are crammed into the same session.[10] Your VO_2max, meanwhile, improves regardless of whether you lift before or after you run.[11]

How to keep interference small

- **Separate the hardest sessions.** Leaving several hours, or putting them on different days, lets each adaptation proceed cleanly.[12]
- **Use intervals as your cardio.** High intensity interval work as your endurance modality does not appear to blunt strength, and it is the most time efficient way to build the engine.[12]
- **Mind the order when it counts.** For longer blocks focused on lower body strength, doing strength before endurance tends to protect strength gains. For VO_2max itself, order does not matter.[11]
- **Periodize.** Shift emphasis across the year rather than maxing everything at once.

The secret weapon: strength builds running economy

Here is the part hybrid athletes should love. **Heavy strength training makes you a more economical runner.** A meta-analysis found heavy resistance work improved running economy more than plyometrics, and that heavy lifting is especially effective at faster running speeds and in athletes who already have a well developed VO_2max.[13] Combining heavy lifting with some plyometric work gave the biggest economy boost of all.[14] So the strength you build for the sled and the wall ball is also quietly making your runs cheaper.

Hyrox specifically: why the engine wins races

The first scientific study of Hyrox confirmed what athletes feel: it is a running race with strength stations bolted on, eight kilometres of running broken up by functional work.[15] A high VO_2max matters twice over. It lets you run the 1k splits faster, and just as importantly it lets you **recover faster** at and after each station, clearing lactate and catching your breath so you can run again sooner.[16] You cannot hold 100 percent of your ceiling for a whole race, so the higher your ceiling and the larger the share you can sustain, the faster your whole day becomes.

THE HYBRID BUILD ORDER

Build a deep aerobic base and solid relative strength first. Once that foundation is in place, layer in Hyrox specific work to develop the local muscular endurance and the mental grit the format demands.[16]

THE TAKEAWAY

You do not have to choose. Keep your hard runs and hard lifts apart, use intervals for the engine, and lean on heavy strength to make your running cheaper. For Hyrox, **VO_2max is the gift that keeps giving**: faster splits and faster recovery.

07 THE FREE WINS

Training is the main event, but a handful of low cost levers quietly raise your ceiling and help your hard work actually stick. These are the things many athletes ignore and later wish they had not.

Sleep: the adaptation you cannot skip

Your engine is not built during the workout. It is built while you recover, and sleep is when most of that happens. Endurance athletes training hard generally need **eight to ten hours**, more than the standard adult guideline.[17] Skimp on it and your body raises stress hormones, lowers the anabolic hormones that drive repair, and turns the same training into accumulated fatigue rather than fitness.[18] Crucially, research shows that **extending** your sleep helps protect endurance performance.[19] This is the cheapest performance upgrade on earth.

TRY THIS · BANK THE SLEEP

Add 30 minutes a night for two weeks. That is roughly an extra full night of sleep banked, and your hard sessions will land better for it. Consistency of bed and wake times matters as much as total hours.

Heat: altitude style gains without a mountain

Training in the heat, or even passive heat exposure like hot baths and saunas, triggers a fast expansion of your blood plasma volume within the first week, and over several weeks it can nudge up your oxygen carrying haemoglobin mass too.[20] A 2025 study in trained runners found that adding regular hot water immersion raised both VO_2max and oxygen carrying capacity, with no extra running at all.[21] More fluid and more oxygen carriers in the system means more oxygen delivered per beat. It is one of the more exciting low tech levers in endurance science right now.

DO THIS SENSIBLY

Heat is a real stressor. Build up gradually, hydrate well, never combine aggressive heat with a fragile or unwell state, and treat it as a supplement to training, not a replacement. If you have any cardiovascular condition, check with a doctor first.

Nutrition and supplements: signal versus noise

Whole food, enough carbohydrate to fuel your hard sessions, and enough protein to repair will always outrank any pill. On supplements, the honest picture matters:

- **Beetroot and dietary nitrate:** Genuinely useful for some, but oversold for VO_2max. The effect on VO_2max itself is small to negligible.[22] Where it helps is reducing the oxygen cost of submaximal effort and extending time to exhaustion, and the benefit is larger in recreational athletes than in elites, whose bodies are already highly optimised.[23]
- **Iron:** Not a booster, but a non negotiable foundation. Low iron caps oxygen transport directly. If you are often fatigued, especially female or a high volume athlete, ask your doctor to check your levels.

THE TAKEAWAY

Sleep is the free win nobody should skip. Heat exposure is a promising bonus lever when done carefully. Fuel real food first, and treat supplements as small extras, not shortcuts.

08 PUTTING IT TOGETHER

Here is one practical way to assemble everything above into a single week. This is a template for a committed amateur runner or hybrid athlete, not a prescription. **Adapt it to your life, your recovery and your event.**

DAY	FOCUS	SESSION	ZONE
Mon	Recovery	Rest, or easy walk and mobility	Easy
Tue	VO_2max	5 × 3 min hard, 3 min jog recovery	Hard
Wed	Easy + strength	Easy run, then heavy lower body lifting (kept separate)	Easy
Thu	Threshold	5 × 6 min comfortably hard, 1 min jog	Mod
Fri	Easy	Easy run, truly easy, or full rest	Easy
Sat	Hybrid / specific	Hyrox style work, or hills, or tempo	Hard
Sun	Long aerobic	Long easy run to build the base	Easy

Notice the shape: the bulk of the week is easy, with a small number of genuinely hard sessions placed where you can recover around them. The two quality days, one VO_2max and one threshold, do the heavy lifting for your engine. Strength is kept away from your hardest run. The long run quietly builds the durable, peripheral fitness from Section 3.

Five principles to carry with you

- **Intensity is the spark.** Protect one or two truly hard sessions a week, structured as 3 to 5 minute reps.
- **Easy means easy.** The base only works if you keep it gentle enough to recover from.
- **Consistency beats intensity over time.** The engine compounds for years. Keep showing up.
- **Strength is not the enemy.** It makes you more economical and more durable.
- **Recover on purpose.** Sleep, fuel and patience turn training stress into fitness.

> **The best plan is not the hardest one. It is the one you can repeat, week after week, for years.**

THE TAKEAWAY

Build your week around a big easy base, one or two quality days, and deliberate recovery. Do that consistently and your engine has no choice but to grow.

NOW GO BUILD THAT ENGINE.

You now know more about raising your VO_2max than most athletes ever will. The science is on your side, your genetics are not your ceiling, and the levers that matter are within your reach: **smart intervals, an easy aerobic base, a little heavy strength, and real recovery.**

Start with one change this week. Then another. The athletes who transform are simply the ones who keep going.

Train with structure, tools and a community that gets it.

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Training plans, free tools and resources for runners and hybrid athletes. Share this guide with someone who needs a bigger engine.

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